

Raul Taranco

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Computer architecture researcher at Huawei's Von Neumann Research Center in Zurich. I model where time and energy go in real workloads and reduce that cost through hardware-software co-design. Ph.D., 2024.

Experience

Researcher, Computer Architecture 2025 – Present

Huawei, Von Neumann Research Center, Zurich, Switzerland

- Built Aneto (**MICRO'26**), a CPU performance model for new memory such as HBM or CXL that uses one run's hardware counters. Selected for early-stage design exploration of Huawei's next-generation CPU.
- Evaluated on 100+ workloads on AMD Zen 2–5, Intel and Arm servers: 12.7% median error against real HBM on Arm. In ChampSim, 2.8× lower p90 error than the best prior one-shot predictor.
- Sped up Arm SME GEMM kernels 1.4× (36 cores) in Huawei's production BLAS library: parallelized a reduction step and fixed cache and prefetch issues. Huawei Director's Award.
- Build LLM-agent workflows that modify and calibrate architecture simulators and assist RTL design.

Research Collaborator 2025 – Present

ARCO Group, UPC, Barcelona, Spain

- Co-supervise the Ph.D. research of Ali Farahani on ray-tracing hardware with Profs. Antonio González and Juan Luis Aragón.

AI Committee Member (part-time) 2024 – Present

Medidedalia, Barcelona, Spain

Postdoctoral Researcher 2024 – 2025

ARCO Group, UPC, Barcelona, Spain

- Built MVP (**MICRO'26**), which predicts motion in the ISP to extrapolate detections between frames. Up to 66.8% lower tail latency and 46% lower energy at a 1.3–1.6% mAP cost, validated in RTL.

Visiting Researcher 2023

Edge Computing Lab, Harvard University, Boston, MA, USA

- Studied computing systems for robotics and autonomous driving, hosted by Prof. Vijay Janapa Reddi.

Ph.D. Researcher (FPU Fellow) 2019 – 2024

ARCO Group, UPC, Barcelona, Spain

- Built δ LTA (**MICRO'23**) and IRIS (**HPCA'25**), camera and ISP co-designs that skip redundant work. They cut average latency by 15–20% and energy by 16–17%.
- Designed SLIDEX (**ICS'24**), a CPU vector extension for sliding-window image processing, with a 1.2× speedup and up to 19% lower energy.
- Designed LOCATOR (**SBAC-PAD'21**, **JPDC'23**), an ORB feature-extraction accelerator for visual localization, with 16.8× faster feature extraction and 2.1× lower energy per frame than a mobile CPU.

Education

Ph.D. in Computer Architecture (Cum Laude), UPC 2024

- **Thesis:** “Architectural Strategies to Enhance the Latency and Energy Efficiency of Mobile Continuous Visual Localization Systems.” **Advisors:** Prof. Antonio González, Dr. José-María Arnau

M.S. in Computer Engineering and High Performance Computing, UPC 2019

B.S. in Computer Science, with Honors, Universidad de Cantabria 2016

Publications

- [1] **Raul Taranco**, Rene Mueller, and Michael Giardino. “[Aneto: Predicting System Performance by Exploiting Cross-Workload Regularity](#)”. In: *Proceedings of the 59th IEEE/ACM International Symposium on Microarchitecture (MICRO’26)*. 2026. Artifact badges: Available, Functional, Reproduced.
- [2] **Raul Taranco** and Antonio González. “MVP: A Motion-Predictive Speculative Vision Pipeline with Non-Blocking Drift Correction”. In: *Proceedings of the 59th IEEE/ACM International Symposium on Microarchitecture (MICRO’26)*. 2026.
- [3] **Raul Taranco**, José-María Arnau, and Antonio González. “[IRIS: Unleashing ISP-Software Cooperation to Optimize the Machine Vision Pipeline](#)”. In: *Proceedings of the 31st IEEE International Symposium on High-Performance Computer Architecture (HPCA’25)*. 2025, pp. 231–245. HiPEAC Paper Award.
- [4] **Raul Taranco**, José-María Arnau, and Antonio González. “[SLIDEX: A Novel Architecture for Sliding Window Processing](#)”. In: *Proceedings of the 38th ACM International Conference on Supercomputing (ICS’24)*. 2024, pp. 312–323.
- [5] **Raul Taranco**, José-María Arnau, and Antonio González. “[δLTA: Decoupling Camera Sampling from Processing to Avoid Redundant Computations in the Vision Pipeline](#)”. In: *Proceedings of the 56th IEEE/ACM International Symposium on Microarchitecture (MICRO’23)*. 2023, pp. 1029–1043. HiPEAC Paper Award.
- [6] **Raul Taranco**, José-María Arnau, and Antonio González. “[SLIDEX: Sliding Window Extension for Image Processing](#)”. In: *Proceedings of the 32nd International Conference on Parallel Architectures and Compilation Techniques (PACT’23)*. 2023, pp. 332–334. Short paper and poster.
- [7] **Raul Taranco**, José-María Arnau, and Antonio González. “[LOCATOR: Low-Power ORB Accelerator for Autonomous Cars](#)”. In: *Journal of Parallel and Distributed Computing (JPDC)* 174 (2023), pp. 32–45.
- [8] **Raul Taranco**, José-María Arnau, and Antonio González. “[Sliding Window Support for Image Processing in Autonomous Vehicles](#)”. In: *Workshop on Compute Platforms for Autonomous Vehicles (CAV’22)*, held with MICRO’22. 2022.
- [9] **Raul Taranco**, José-María Arnau, and Antonio González. “[A Low-Power Hardware Accelerator for ORB Feature Extraction in Self-Driving Cars](#)”. In: *Proceedings of the 33rd IEEE International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD’21)*. 2021, pp. 11–21.

Awards

- **2026:** Huawei Director’s Award (Timely Incentives) for SME kernel work in the production BLAS library
- **2025, 2023:** HiPEAC Paper Awards for **HPCA’25** (IRIS) and **MICRO’23** (δLTA)
- **2019:** FPU Ph.D. Fellowship (Spanish Ministry of Education)
- **2017:** B.S. Extraordinary Award (top graduate), *Universidad de Cantabria*

Service

- **Program Committee:** MICRO’26, HPCA’27
- **Journal Reviewer:** IEEE Computer Architecture Letters (CAL), IEEE Access

Skills

- **Programming:** C/C++, Python, Bash, OpenMP, MPI, Arm SME
- **Modeling and simulation:** gem5, ChampSim, Sniper, DRAMSim, McPAT, Linux perf, PMU counters
- **Hardware design:** Verilog, VHDL, Synopsys Design Compiler, Yosys
- **Languages:** Spanish (native), English (fluent)